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PHASE DIAGRAMS OF THE CaF_2–(Y, Ln)F_3 SYSTEMS
I. EXPERIMENTAL

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Summary

The phase diagrams of the CaF_2–(Y, Ln)F_3 systems, where Ln is any one of the lanthanides except Pm and Eu, were studied by thermal and X-ray phase analysis methods. The investigations were made from 800 (600) °C up to the melting temperature. The oxygen concentration of the specimens was checked after thermal treatment. For the systems with Ln = Ce, Pr, Tb, Ho, Tm and Lu the fusibility data have been obtained for the first time. Phases with the following structure types are present in the systems investigated: (a) solid solutions Ca_{1-x}(Y, Ln)$_x F_{2+x}$ with the fluorite structure type, which undergo ordering in the systems with Ln = Ho - Lu, Y; (b) non-stoichiometric phases (Y, Ln)$_{1-y}Ca_yF_{3-y}$ with the tysonite (LaF_3) structure type, which undergo ordering in the systems with Ln = Tb - Lu, Y; (c) solid solutions with the α-UO_3 structure type, based on α modifications (Y, Ln)F_3 (Ln = Er - Lu); and (d) modifications (Y, Ln)F_3 (Ln = Gd - Lu) of the β-YF_3 type. Successive transitions, from solid solutions of the components of the system to the berthollide phases and the stoichiometric compound, were found for phases with the tysonite structure type. The maxima on fusion curves of non-stoichiometric phases were noticed in Sm, Gd, Tb, Ho and Er for Ca_{1-x}-(Y, Ln)$_x F_{2-x}$ and for Ln = Pr - Lu in (Y, Ln)$_{1-y}Ca_yF_{3-y}$. The coordinates of non-variant points for equilibria of the liquid–solid type and also the equations describing the dependence of the lattice parameters of non-stoichiometric phases on the composition are given for the systems investigated.

1. Introduction

The history of the experimental investigation of CaF_2–(Y, Ln)F_3 systems (Ln = lanthanum and lanthanides) begins in 1914 [1], when the fusion of YF_3 solid solutions in fluorite was found to have a maximum at about 12 mol.% of yttrium fluoride.

Interest in these systems (or, more precisely, in $Ca_{1-x}Ln_xF_{2+x}$ solid solutions with the fluorite structure type) grew after Goldschmidt [2]

proposed a scheme of heterovalent isomorphous substitutions of Ca^{2+} by Ln^{3+} in the fluorite structure type. This scheme envisaged valency compensation by additional interstitial fluoride ions in the centres of the empty cubes of the fluorite structure and offered a basis for the isolation of a new type of "isomorphism with the occupation of space" in ionic crystals. To test this hypothesis, measurements of the density of solid solutions of the fluorite structure type were made and compared with the theoretically calculated values for various models [3 - 14]. Most of the measurements provided support for the model described in ref. 2, but structural studies of $Ca_{0.607}Ce_{0.393}F_{2.393}$ [15] and $Ca_{0.9}Y_{0.1}F_{2.1}$ [16] showed that the positions of additional fluorine atoms did not conform to the proposal advanced by Goldschmidt. The mechanism of defect interaction in such solid solutions has recently been discussed [17].

The distinctive characteristics of the crystalline structure of solid solutions $Ca_{1-x}R_xF_{2+x}$ with the fluorite structure type find expression in the peculiar spectroscopic behaviour of ion activators in multicomponent matrices [18]. After the production of the first laser on fluorides ($CaF_2:U^{3+}$ [19]) a large number of mixed materials for lasers based on MF_2 with the fluorite structure type were obtained. A detailed characterization of these laser materials is available [20].

The growth of single crystal materials based on binary (and more complex) systems of MF_2–(Y, Ln)F_3 type aroused interest in the investigation of their phase diagrams. In 1963 the CaF_2–YF_3 system was studied for the first time after an interval of about 50 years [8]. The study has since been repeated [21, 22]. The phase diagrams of the following systems have now been published: CaF_2–LaF_3 [23, 24], CaF_2–NdF_3 [21, 23], CaF_2–SmF_3 [25], CaF_2–GdF_3 [26], CaF_2–DyF_3 [25], CaF_2–ErF_3 and CaF_2–YbF_3 [27]. A very large number of papers describe the synthesis of single crystals from various ranges of composition of the CaF_2–(Y, Ln)F_3 systems and the study of their physical characteristics. We shall mention those papers which contain information about phase equilibria in the CaF_2–LnF_3 systems.

Of the important fields of use of materials obtained on the basis of CaF_2–LnF_3 systems, we must mention materials for the optical processing of information [28 - 30] and the conversion of IR quantums into visible light [31]; the components and certain compositions of systems are used as solid electrolytes with high ionic conductivity [32, 33], thin film condensers [34] etc. A pronounced effect of Ln^{3+} impurities on the mechanical characteristics of fluorite [35] was noted. The fusibility of these systems is of interest for an understanding of the electroslag melting of certain metals [24] and the physicochemical basis of the process of thermal reduction of lanthanide fluorides by calcium [36].

Despite the fact that numerous investigations of systems of the CaF_2–(Y, Ln)F_3 type (and their components) have been made, a large number of problems still remain unsolved. For the construction of correct phase diagrams it is necessary to know the temperatures of all phase transitions of the components within the investigated temperature range. The information

available for trifluoride lanthanides was very contradictory. We have investigated [37, 38] the effect of heterovalent anion (oxygen) and cation (Ca, Sr, Ba) impurities on the temperature of phase transformations in LnF_3. This effect was very pronounced. For this reason the earlier studies of systems containing LnF_3 are partially invalidated, because measurements were not made of the oxygen concentration of the specimens after thermal treatment. In certain cases [12, 26] it is clear that the pyrohydrolysis of the specimens was significant and that the data obtained were incorrect.

The results of investigations of phase equilibria in the CaF_2–(Y, Ln)F_3 systems made by annealing and quenching methods [39, 40], growth of single crystals [14] and thermal analysis [25 - 27] lead to significantly different results, particularly in systems containing heavy lanthanide trifluorides. For example, the liquidus curves of solid solutions with the fluorite structure type in the CaF_2–SmF_3 and CaF_2–GdF_3 systems [25, 26] do not have maxima. This does not correlate with data for the distribution coefficients of Ln^{3+} in CaF_2 [41 - 43]. Such discrepancies are serious and relate not only to fusibility but also to the structural types of intermediate phases and their existence in certain systems.

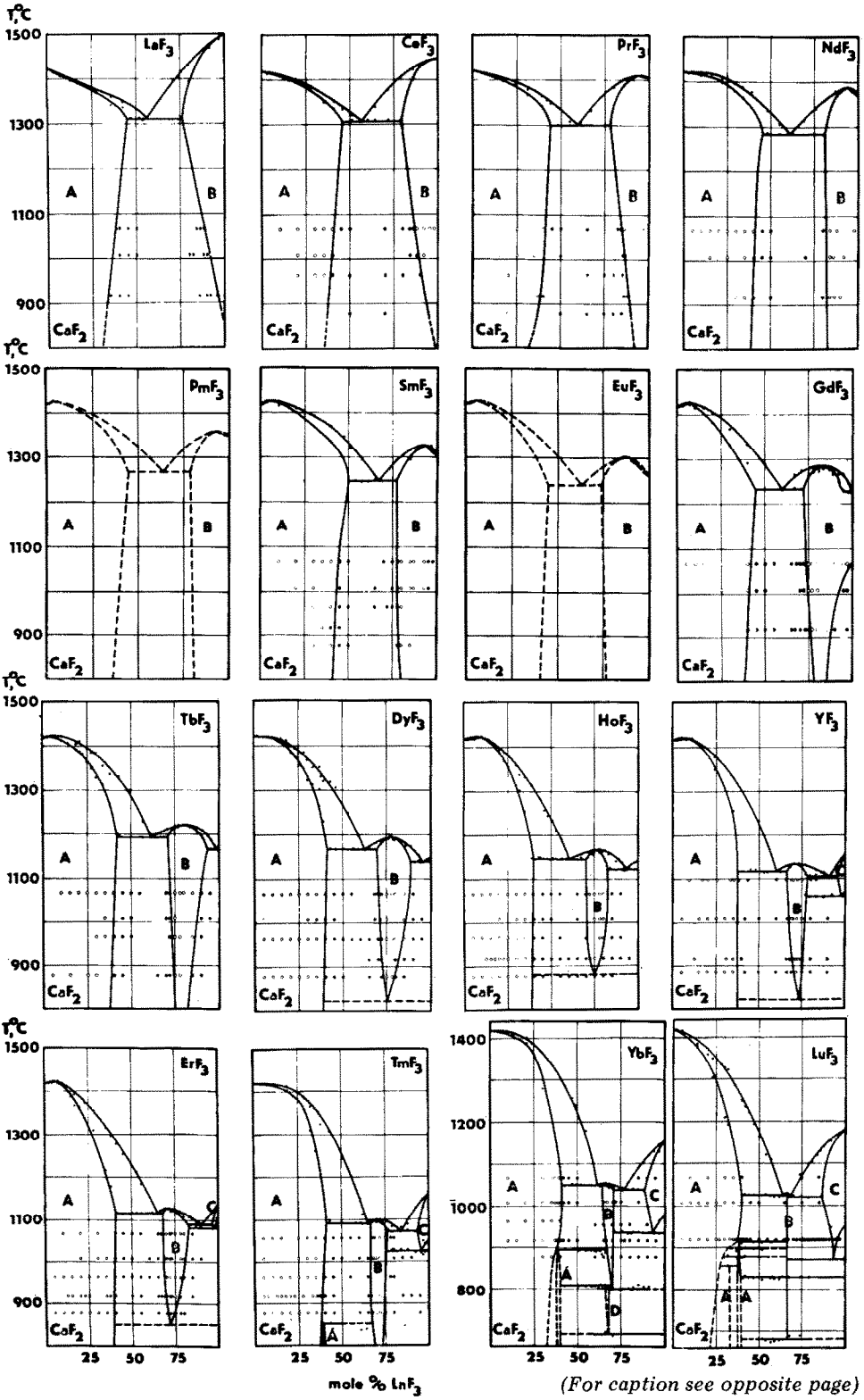
In our opinion, some of these discrepancies are due to the distinguishing characteristics of the thermal behaviour of phases in the CaF_2–(Y, Ln)F_3 systems. Thus the decomposition of solid solutions $Ca_{1-x}Ln_xF_{2+x}$ with the fluorite structure type is a very slow process. For this reason prolonged annealing is necessary to achieve states of equilibrium in these ranges of composition. In the (Y, Ln)F_3-rich composition range, phases are formed which are not quenched at all or are partially quenched (polymorphous modifications of LnF_3 and their solid solutions, berthollide phases). For the study of these transitions, dynamic methods, such as thermal and high temperature X-ray analysis, are necessary. Thus the construction of complete phase diagrams for the CaF_2–(Y, Ln)F_3 systems requires the use of various methods of investigation.

The present paper describes systematically the information obtained by us about the phase diagrams of the CaF_2–(Y, Ln)F_3 systems.

2. Experimental procedure

Optical quality single crystals of CaF_2 (oxygen concentration 0.025 wt.%) and trifluoride lanthanides obtained from the oxides (grades 00 and 000) fused in a fluorinating atmosphere (oxygen concentration 0.005 - 0.07 wt.%) were used as starting reagents. The concentration of oxygen was determined by the vacuum fusion method [44]; the experimental error was ±25%.

The procedures employed in the thermal analysis, the study of equilibria in the subsolidus parts of the phase diagrams and the X-ray phase analysis have been described previously [22, 37, 45, 46]. The oxygen concentration in the specimens after thermal analysis was on average 0.04%; after sintering in the solid phase it varied from 0.07 to 0.15 wt.%.



(For caption see opposite page)

The thermocouples W–5%Re/W–20%Re were calibrated by the melting points of CaF_2 (1418 °C [47 - 51]) and YF_3 (1152 °C [52 - 55]). The average error was ±10 °C in the temperature determination and ±3 °C in the reproducibility. The tendency of a large number of compositions of the CaF_2–(Y, Ln)F_3 systems to supercooling of melts calls for the use of heating curves to detect phase transformations. This tendency has been observed before [37, 56].

3. Structural types and crystallographic characteristics of phases appearing in the CaF_2–(Y, Ln)F_3 systems

The results of the study of phase diagrams of the CaF_2–(Y, Ln)F_3 systems are presented in Fig. 1. The coordinates of non-variant points are given in Table 1. Of the sixteen diagrams, fourteen were studied experimentally. The CaF_2– PmF_3 and CaF_2–EuF_3 phase diagrams* were constructed by the interpolation of variations in non-variant point coordinates for a series of lanthanides. The interpolation accuracy was close to the experimental error. We have previously published some phase diagrams of the CaF_2–(Y, Ln)F_3 systems and also brief information about the experimental results [22, 38, 46, 58 - 60].

3.1. Solid solutions with the fluorite structure type

These solutions (Fig. 1, phase A) are formed in all systems. They are characterized by the unordered distribution of Ca^{2+} and Ln^{3+} and retain the space group $Fm3m$ [15, 16]. The relationship between the parameters of elemental cells and the composition of equilibrated specimens shows deviations from Vegard's rule at high concentrations of LnF_3. Such deviations are not observed if the specimens are obtained by quenching from the melt (see, for example, refs. 8, 39, 61 - 63). The coefficients of equations (see Table 2) describing the relationships shown in Fig. 2 were calculated by the least squares method. In Table 2 the homogeneity areas (in terms of composition) of saturated unordered solid solutions $Ca_{1-x}Ln_xF_{2+x}$ with the fluorite structure type at eutectic temperatures are also given. It must be pointed out that the solubility limits of LnF_3 in CaF_2 are very large and do not vary significantly on decreasing the ionic radius by 15% from La^{3+} to Lu^{3+}. The temperature dependence of solubility is clearly defined in systems containing the trifluorides of light lanthanides, but with increasing molecular weight of

*It was revealed by magnetochemical measurements that europium is partially reduced under our experimental conditions [57].

Fig. 1. Phase diagrams of the CaF_2–(Y, Ln)F_3 systems: ●, thermal analysis data; ○, single phase specimens; ◑, two-phase specimens.

TABLE 1

Coordinates of non-variant points of the liquid–solid equilibrium and the eutectoid in the CaF_2–(Y, Ln)F_3 systems (c, mol.% RF_3; T, °C)

RF_3	Maxima on fusibility curves of cubic solid solutions		Eutectic point I		Maxima on fusibility curves of tysonite-type solid solutions		Eutectic point II		Eutectoid (solid solution with α-YF_3 structure type)	
	c	T	c	T	c	T	c	T	c	T
LaF_3	–	–	58	1311	–	–	–	–	–	–
CeF_3	–	–	57	1308	–	–	–	–	–	–
PrF_3	–	–	59	1300	95	1410	–	–	–	–
NdF_3	–	–	60	1284	94	1386	–	–	–	–
(PmF_3)	(3)	(1425)	(64)	(1266)	(93)	(1360)	–	–	–	–
SmF_3	5	1428	67	1248	92	1327	–	–	–	–
(EuF_3)	(5)	(1428)	(63)	(1240)	(88)	(1300)	–	–	–	–
GdF_3	5	1428	60	1233	85	1282	–	–	–	–
TbF_3	4	1420	61	1194	81	1222	99	1166	–	–
DyF_3	–	–	62	1168	78	1194	96	1139	–	–
HoF_3	10	1424	60	1147	77	1163	93	1122	–	–
ErF_3	7	1423	62	1117	70	1128	90	1090	98	1082
TmF_3	–	–	65	1092	70	1095	86	1071	95	1023
YbF_3	–	–	61	1049	67	1053	77	1040	94	937
LuF_3	–	–	63	1025	67	1027	69	1023	93	872
YF_3	11	1422	60	1120	71	1135	91	1106	98	1063

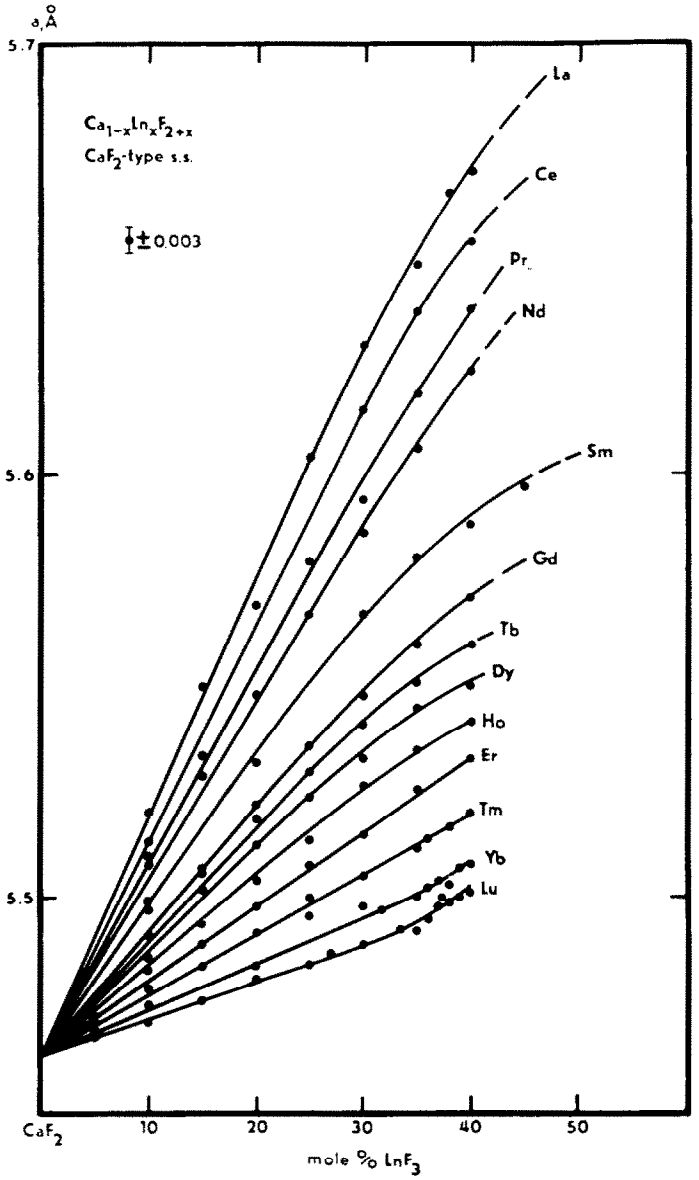


Fig. 2. Graph of the parameters of unit cells of the $Ca_{1-x}Ln_xF_{2+x}$ phases *vs.* the concentration of LnF_3.

LnF_3 this effect diminishes. In CaF_2 systems with YbF_3 and LuF_3, the range of existence of unordered $Ca_{1-x}Ln_xF_{2+x}$ solid solutions is restricted by the separation of ordered phases. The heterovalent isomorphous substitution of Ca^{2+} by Ln^{3+}, which has been demonstrated by structural investigations [15, 16], is followed by the formation of various types of structural defects. Besides the substitution defects in cation sites, vacancies are formed in the

TABLE 2

Coefficients of the equations describing the relationship between the parameters of unit cells of $Ca_{1-x}Ln_xF_{2+x}$ phases with the fluorite structure type and their composition

R	Homogeneity area x	k_1	k_2	Dispersion[a] $S^2 \times 10^6$
La	0 - 0.46	0.597	0.134	0.7
Ce	0 - 0.45	0.562	0.197	11.0
Pr	0 - 0.43	0.483	0.113	3.7
Nd	0 - 0.44	0.455	0.132	0.8
Sm	0 - 0.50	0.375	0.142	1.3
Gd	0 - 0.45	0.320	0.121	5.6
Tb	0 - 0.42	0.310	0.164	0.9
Dy	0 - 0.41	0.270	0.121	5.5
Ho	0 - 0.40	0.215	0.029	5.0
Er	0 - 0.40	0.176	–	2.4
Tm	0 - 0.40	0.144	–	3.1
Yb	0 - 0.40	0.112	–	15.3
Lu	0 - 0.40	0.088	–	2.4
Y	0 - 0.38	0.191	–	3.7

a (Å) = $a_0 + k_1x + k_2x^2$; a_0 = 5.46295 Å [64].
[a]The dispersion calculation is described in ref. 46.

basic anion sublattice (V_F^-); other defects are due to interstitial fluorine atoms (F_i^-). In the $Ca_{0.9}Y_{0.1}F_{2.1}$ solid solution two sites of interstitial fluorine atoms have been identified [16]: $F_{i,1}^-$ and $F_{i,2}^-$. The total number of fluorine atoms in the inter-site positions is equivalent to a double molar fraction (x) of RF_3 in the $Ca_{1-x}R_xF_{2+x}$ solid solution [15, 16]. The number of vacancies in the basic anion sublattice V_F^- is equivalent to the molar fraction of RF_3. Taking this into account and using the Kröger–Vink symbols for defects, the scheme of heterovalent isomorphous substitutions in the fluorite structure type which takes account of quantitative interaction between the defects should be written as

$$Ca^{2+} + F_{basis}^- \rightarrow R_{Ca}^{3+} + mF_{i,1}^- + nF_{i,2}^- + V_F^- \quad (1)$$

where $m + n = 2$ and the distribution of interstitial atoms F_i^- in the positions (i, 1) and (i, 2) depends upon the RF_3 concentration.

The formation of $Ca_{1-x}R_xF_{2+x}$ solid solutions takes place with an increase in the degree of space occupation and an appreciable dependence of many physical characteristics of the $Ca_{1-x}R_xF_{2+x}$ phases on the concentration. This is an important condition for obtaining materials with the required properties.

The $Ca_{1-x}R_xF_{2+x}$ solid solutions show clear evidence of the tendency of fluorite-type structures to give rise to non-stoichiometric phases with a high concentration of structural defects.

3.2. Ordered phases with a CaF_2-derived structure type

The ordering phenomenon of non-stoichiometric phases with the fluorite structure type for three series of MF_2–(Y, Ln)F_3 systems where M = Ca [65], Sr [66] or Ba [67] was discovered and described for the first time by us.

In CaF_2–(Y, Ln)F_3 systems where Ln = Ho - Lu, the X-ray study of single crystals of these phases reveals a 13-fold increase in the parameters of the unit cells; this, together with the concentration ranges in which they exit, suggests an idealized composition of $Ca_8R_5F_{31}$ [65]. In Fig. 1 these phases are denoted as A′. In the systems with HoF_3, ErF_3 and YF_3, ordering takes place at 690 - 750 °C and is not plotted in Fig. 1. In the system with LuF_3 the $Ca_8Lu_5F_{31}$ phase apparently undergoes two polymorphous transitions and another ordered phase A″ of approximate composition Ca_2LuF_7 appears. This area of the phase diagram is incomplete.

The thermal effects appropriate to ordering have a noticeable temperature hysteresis; this indicates that the order–disorder transition cannot be a second order transition. This is not in agreement with our previous observations [65], which could be interpreted in terms of a concentrating-type continuous transition from the disordered to the ordered phase. Apparently the partially ordered specimens for which power X-ray diagrams and Laue photographs have been obtained [65] remained in the metastable states which appear upon quenching the high temperature unordered solid solutions.

It is necessary to point out that the discrepancy between the observed cubic symmetry and the slight rhombohedral distortion of unit cells of the $Ca_8R_5F_{31}$ phases [65] is analogous to the behaviour of the U_4O_9 phase ordered with respect to the solid solution with the CaF_2 structure type [68]. The rhombohedral distortion of the ordered cubic phase on lowering the temperature is accompanied by twinning of rhombohedral domains along four triad axes of the high temperature phase; the elastic stresses ensure their coherence and imitation by the cubic symmetry. This leads us to assume the presence of ferroelectrical transformations in $Ca_8R_5F_{31}$ [69].

3.3. Solid solutions with the LaF_3 structure type

These solutions are also formed in all the systems studied (phase B in Fig. 1). According to the powder X-ray data [40] the $Ln_{1-y}Ca_yF_{3-y}$ phases are apparently characterized by the unordered distribution of Ln^{3+} and Ca^{2+} in the cation sites of the LaF_3 structure type [70, 71]. The study of the dependence of the parameters of the hexagonal unit cell on the composition [60, 72] did not reveal any deviations from Vegard's rule for annealed and quenched specimens of the majority of the systems (except phases with La, Ce, Pr, Nd); similarly, no singular points were found to be associated with the maxima on the fusibility curves of $Ln_{1-y}Ca_yF_{3-y}$ phases (Fig. 1). The coefficients of equations describing the dependence of the unit cell parameters of $Ln_{1-y}Ca_yF_{3-y}$ phases on the concentration are given in Table 3. The measurement of the densities of the solid solutions $Ln_{1-y}Ca_y$-

TABLE 3

Coefficients of the equations describing the concentration dependence of parameters of unit cells of tysonite-type phases $(Y, Ln)_{1-y}Ca_yF_{3-y}$

R	a (Å) $= a_0 + k_1y + k_2y^2$				c (Å) $= c_0 + k_3y$			Homogeneity area y
	a_0	$-k_1$	$-k_2$	$S^2 \times 10^6$[a]	c_0	$-k_3$	$S^2 \times 10^6$[a]	
La	7.186[b]	0.256	1.376	20.4	7.352[b]	0.167	11.8	0 - 0.23
Ce	7.125[b]	0.235	0.912	5.4	7.292[b]	0.199	17.8	0 - 0.21
Pr	7.078[b]	0.149	1.312	7.0	7.239[b]	0.099	6.4	0 - 0.22
Nd	7.030[b]	0.172	1.142	3.0	7.200[b]	0.035	21.6	0 - 0.19
Sm	6.966	0.353	–	30.9	7.136	0.195	26.0	0 - 0.24
Gd	6.893	0.118	–	33.2	7.071	–	12.3	0 - 0.30
Tb	6.864	0.136	–	4.3	7.028	–	26.5	0.05 - 0.29
Dy	6.831	0.063	–	17.5	7.001	–	15.4	0.09 - 0.30
Ho	6.807	0.082	–	1.3	6.975	–	16.6	0.14 - 0.30
Er	6.798	0.105	–	7.5	6.952	–	5.2	0.17 - 0.33
Tm	6.752	–	–	–	6.928	–	–	0.24 - 0.33
Yb	6.735	–	–	–	6.908	–	–	0.28 - 0.35
Lu	6.713	–	–	–	6.906	–	–	0.33
Y	6.786	–	–	0.8	6.960	–	0.2	0.22 - 0.34

[a] The dispersion calculation is described in ref. 46.
[b] The values a_0 and c_0 were pre-assigned and taken from ref. 72.

F_{3-y} [21, 40] favours the substitution of Ln^{3+} by Ca^{2+} with the formation of vacancies V_F^- in the anion sublattice. The densities of the non-stoichiometric phases with the compositions $Y_{0.7}Ca_{0.3}F_{2.7}$ and $Er_{0.725}Ca_{0.275}F_{2.725}$ were 4.48 and 6.56 g cm^{-3}. This is also in agreement with the theoretically calculated densities (4.51 and 6.64 g cm^{-3}) for the anion vacancies model:

$$Ln^{3+} + F^- \longrightarrow Ca_{Ln}^{2+} + V_F^- \qquad (2)$$

In the annealed and quenched specimens of the CaF_2–(Y, Ln)F_3 systems where Ln = Tb - Yb, and also in the specimens quenched from the melt, phases with the tysonite-derived structure described earlier were found by the X-ray powder method [40, 62]. The strictly defined concentration areas in which these phases were found led us to a conclusion about the concentrated nature of ordering phases with the tysonite structure type [40]. In accordance with this we interpreted the phase equilibria in the CaF_2–YF_3 system [22].

However, during the thermal analysis, low temperature endothermic effects were noticed by us in the phase diagrams. By the high temperature X-ray method we succeeded in proving for the $Tm_{0.7}Ca_{0.3}F_{2.7}$ specimen that the effect at 700 °C conforms to the reversible transition of the ordered phases into unordered phases with the LaF_3 structure type. The new interpretation of interrelationships of phases with the tysonite (B) structure type and the derivatives of its structural type (D) is shown in Fig. 1 for the CaF_2–YbF_3 system. The problem of the interrelationships of these phases and their position on the phase diagrams needs additional study. The atomic structure of the ordered phase is of much interest.

We have shown that the distinguishing characteristics of the tysonite structure type determine the upper limit ($CaLn_2F_8$ composition) of isomorphous substitution of Ln^{3+} by Ca^{2+} [60]. The $CaLn_2F_8$ composition is obtained in the CaF_2–LuF_3 system. In this case all the F^{III}-type [70] atoms which are loosely bonded to the cations (amounting to 2/18 of the fluorine atoms in the elemental cell with $z = 6$) are removed from the structure. The homogeneity area of $Ln_{1-y}Ca_yF_{3-y}$ phases decreases with the increase of the lanthanide atomic number (see Fig. 1 and Table 3) and shrinks towards the crystal chemical limit of $CaLn_2F_8$. This tendency is related to the change in the electronic structure of Ln^{3+} and its coordinating capacity [73].

3.4. Solid solutions with the α-YF_3(α-UO_3) structure type

These are formed on the basis of high temperature α-LnF_3 modifications which crystallize in this structural type [74 - 77] (phase C in Fig. 1). We did not succeed in obtaining the X-ray characteristics of α-$(Ln_{1-y}Ca_y)F_{3-y}$ phases in any of the systems because they are not quenched.

The isostructural α-UO_3 phases were quenched [78] in the CaF_2–TlF_3 and SrF_2–TlF_3 systems. In the CaF_2–LnF_3 systems with Ln = Tm - Lu, the formation of isotype phases is observed on quenching the compositions close to LnF_3. The powder X-ray diagrams of these phases have broad reflections. In the specimen of composition $Lu_{0.95}Ca_{0.05}F_{2.95}$ annealed at 917 °C for 370 h

the following typical interplanar spacings were observed: 4.171 (60), 3.678 (20), 3.596 (90), 3.461 (30), 3.401 (85), 3.182 (100) and 2.838 (65). Relative intensities are shown in parentheses. We have not yet succeeded in identifying these phases.

3.5. Phases with the orthorhombic β-YF_3 structure type

These phases [79, 80] (orthorhombic trifluorides of Gd - Lu and YF_3) are components of the CaF_2–LnF_3 systems. As far as the accuracy of X-ray analysis permits us to judge, they do not dissolve CaF_2. However, upon quenching specimens of the CaF_2–SmF_3 and CaF_2–GdF_3 systems at temperatures higher than the phase transformation temperatures $\alpha \rightleftharpoons \beta$-$LnF_3$, metastable solid solutions with the β-YF_3 structure type were obtained, containing 5 and 4 mol.% CaF_2, respectively. For the solid solution $Gd_{1-y}Ca_yF_{3-y}$ the molecular volume increases from 48.5 to 50.0 Å when the transition from the tysonite structure type to β-YF_3 takes place.

In the present paper we describe the experimental data on the study of phase diagrams of fourteen CaF_2–(Y, Ln)F_3 binary systems. On the scale of representation selected for Fig. 1 we could not show all the phase fields for certain systems. This will be done by considering the general trends of phase diagrams.

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